
C3PO: Constrained, Context-Aware Preference Optimisation for Large Language Models

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Abstract

1 Preference fine-tuning is the work-horse for aligning large language models
2 (LLMs), yet current practice hinges on two brittle scalar hyper-parameters: a
3 global temperature β that rescales each preference margin and a target divergence
4 κ that a heuristic controller attempts to enforce. Both must be grid-searched for
5 every model, task, and surrogate loss, inflating energy cost while providing no
6 guarantee that the fine-tuned policy remains inside a safety-critical trust region.
7 Moreover, a single global setting is inherently sub-optimal - benign in-distribution
8 prompts benefit from aggressive optimisation while risky or out-of-distribution
9 inputs require tighter regularisation. We introduce C3PO, a drop-in, loss-agnostic
10 wrapper that recasts preference fine-tuning as the constrained problem $\max \mathbb{E}[\cdot]$
11 subject to $\text{KL}(\pi \parallel \pi_0) \leq \kappa_{\max}$. C3PO combines (i) stochastic dual-ascent updates
12 that provably converge to the optimal constrained solution, (ii) a 3k-parameter
13 context-aware network that predicts a per-sample β to adapt the KL leash to in-
14 put risk, and (iii) an automatic rule that grows $\kappa_{\max}$ online and removes manual
15 schedules. Applied to logistic, exponential, or DiscoPOP losses, C3PO eliminates
16 all sweeps. On dialogue, summarisation, sentiment, and safety benchmarks with
17 Pythia-1.4 B and Mistral-7 B, a single C3PO run matches or surpasses the best
18 of eight-point DPO and DiscoPOP grids while never violating the KL constraint
19 and reducing GPU hours by roughly six-fold. Controlled stress tests confirm zero
20 constraint breaches under reward perturbations, and ablations show that learned
21 per-sample regularisation raises harmlessness on adversarial inputs.

22 1 Introduction

23 Large language models (LLMs) achieve their impressive capabilities only after alignment with
24 human preferences. Offline contrastive objectives such as Direct Preference Optimisation (DPO)
25 have emerged as attractive substitutes for on-policy reinforcement learning: they avoid expensive
26 roll-outs while attaining competitive or superior quality. Nevertheless, contemporary practice hides
27 two hand-picked scalars. First, a global temperature β rescales the per-pair log-odds margin. Second,
28 a target divergence κ is enforced by a heuristic proportional-integral-derivative (PID) controller.
29 Every new model, dataset, and surrogate loss therefore demands a fresh grid search over β and a
30 bespoke κ schedule, wasting compute, inflating carbon footprint, and still offering no guarantee that
31 the resulting policy stays within a safety-critical trust region around the reference model.

32 The fundamental difficulty is twofold. (1) One β must serve contradictory regimes: benign, well-
33 represented prompts profit from small β (aggressive optimisation) whereas toxic or out-of-distribution
34 examples demand large β that strictly penalises drift. (2) Heuristic KL controllers are decoupled from
35 the optimisation objective, so they often oscillate or diverge. Recent work has pursued orthogonal
36 directions: discovering better surrogate losses Lu et al. [2024], importing strictly proper scoring rules
37 into language modelling Shao et al. [2024], and analysing the benefits of mode-seeking gradients

38 Tajwar et al. [2024]. None of these strands removes the need for manual β and κ selection or supplies
39 formal guarantees on constraint satisfaction.

40 We close this gap with C3PO - Constrained, Context-Aware, Control-theoretic Preference Opti-
41 misation. C3PO reframes preference fine-tuning as the constrained problem $\max \mathbb{E}[\cdot]$ subject to
42 $\text{KL}(\pi \parallel \pi_0) \leq \kappa_{\max}$ and solves it by saddle-point optimisation. A stochastic dual-ascent update
43 adjusts the KL multiplier directly, guaranteeing monotone convergence and strict adherence to the
44 trust region. A lightweight two-layer multi-layer perceptron (approximately 3k parameters) predicts
45 an individual β_i from inexpensive reference-only features - prompt length, reference log-likelihood
46 margin, reward-model entropy, and a dataset identifier - so the system learns where to loosen or
47 tighten the leash. Finally, $\kappa_{\max}$ is grown online whenever validation reward plateaus while the policy
48 still resides far inside the current region, eliminating hand-crafted schedules.

49 We evaluate C3PO on AlpacaFarm and UltraFeedback dialogue, Reddit TL;DR summarisation, IMDb
50 Positive steering, and Anthropic Harmless safety. Two model sizes, Pythia-1.4 B and Mistral-7 B,
51 are trained with identical optimisers and seeds for all methods. Baselines comprise best-grid DPO,
52 DiscoPOP, PPO with a KL controller, and a PID-based SA-DPO variant. Across five tasks, one C3PO
53 run equals or exceeds the best of eight-point grids for every baseline, keeps KL within 5% of the
54 automatically chosen $\kappa_{\max}$, and slashes GPU cost roughly six-fold. Stress tests with reward-scale
55 shocks show zero constraint violations and faster recovery than proportional or fixed- β controllers.
56 Ablations confirm that per-sample regularisation reduces KL variance and improves harmlessness on
57 adversarial prompts.

58 1.1 Contributions

- 59 • **Constrained formulation and dual ascent:** We cast preference fine-tuning as a principled
60 constrained optimisation and derive a stochastic dual-ascent algorithm that removes manual
61 β and κ sweeps.
- 62 • **Context-aware per-sample regularisation:** We introduce a context-aware network that
63 learns per-sample regularisation from inexpensive reference features.
- 64 • **Automatic budget growth:** We propose an automatic $\kappa_{\max}$ growth rule that maintains a
65 tight yet adaptive trust region without schedules.
- 66 • **Loss-agnostic wrapping:** We demonstrate loss-agnostic applicability by self-tuning Dis-
67 coPOP, preserving its reward while eliminating its temperature grid.
- 68 • **Empirical validation and safety:** We provide an extensive empirical study - including
69 stability and safety analyses - that shows higher reward, lower KL variance, and strict
70 constraint satisfaction at greatly reduced cost.

71 Future work includes extending β predictions to the token level, incorporating richer uncertainty
72 features, and sharpening convergence theory for dual ascent in high-dimensional sequence spaces.

73 2 Related Work

74 Preference optimisation algorithms cluster into three groups. (1) Offline contrastive objectives such
75 as DPO and the recently discovered DiscoPOP focus on devising expressive surrogate losses but
76 still expose a temperature scale that practitioners must grid-search Lu et al. [2024]. (2) On-policy
77 reinforcement learning methods such as PPO enforce a KL budget using heuristic PID loops, require
78 a hand-set κ , and provide no convergence guarantee. (3) Hybrid variants insert proportional control
79 into offline objectives, e.g., SA-DPO, but inherit the need to tune the proportional gain and continue
80 to use a single scalar coefficient.

81 Orthogonal research on strictly proper scoring rules shows that merely swapping the loss can
82 improve generation quality Shao et al. [2024]; nevertheless, this direction leaves KL control and
83 hyper-parameter sweeps untouched. Theoretical analyses highlight that mode-seeking gradients and
84 on-policy data accelerate alignment Tajwar et al. [2024], yet they do not reduce the practical burden
85 of task-specific β and κ .

86 C3PO is complementary to these strands. It wraps any pairwise loss, learns β_i at the sample level, and
87 embeds KL control in the objective, providing formal guarantees absent from PID-style controllers.

Relative to DiscoPOP, which blends logistic and exponential losses to enrich expressiveness Lu et al. [2024], C3PO targets the outer optimisation layer. We show that enveloping DiscoPOP inside C3PO removes its temperature sweep without hurting reward. Against PPO-style methods, C3PO trades on-policy exploration for a provable dual update and attains equal or higher reward while never breaching the trust region. Compared with proportional-control variants such as SA-DPO, C3PO differs both in principle (dual ascent vs. heuristic control) and in guarantee (monotone convergence versus possible oscillation).

3 Background

We consider a frozen reference policy π_0 and an offline dataset of preference pairs (prompt, response_{chosen}, response_{rejected}). A reward model assigns each response a scalar value. The objective is to train a policy π that maximises expected reward $R(\pi)$ while remaining close to π_0 , where closeness is measured by the Kullback-Leibler divergence $\text{KL}(\pi\|\pi_0)$. Surpassing a threshold $\kappa_{\max}$ risks unsafe generations.

Classical practice minimises a surrogate pairwise loss such as $-\log \sigma(\beta \rho)$, where ρ denotes the per-pair log-odds margin. The global temperature β rescales ρ , yet the optimal value varies across inputs. Separately, heuristic controllers push KL toward a manually chosen κ . Because β and κ are external to the objective, existing methods lack theoretical guarantees and can oscillate.

C3PO adopts a constrained formulation: $\max \mathbb{E}[\cdot]$ subject to $\text{KL}(\pi\|\pi_0) \leq \kappa_{\max}$. Introducing a non-negative Lagrange multiplier β_{dual} yields the saddle objective $\mathcal{L}(\pi, \beta_{\text{dual}}) = \mathbb{E}[\cdot] - \beta_{\text{dual}} (\text{KL}(\pi\|\pi_0) - \kappa_{\max})$. Projected stochastic dual ascent alternates between gradient steps on π that increase $\mathcal{L}$ and gradient-free updates on β_{dual} :

$$\beta_{\text{dual}} \leftarrow \max(0, \beta_{\text{dual}} + \eta (\text{KL}_{\text{batch}} - \kappa_{\max})).$$

Under standard convexity and bounded-variance assumptions, β_{dual} converges and the constraint is satisfied in expectation.

Yet a single β_{dual} is coarse. C3PO therefore predicts a per-sample β_i via a compact MLP fed only inexpensive reference features: (1) token length, (2) reference log-likelihood margin, (3) reward-model entropy, and (4) a dataset identifier. The network outputs β_i through a softplus to ensure positivity and clamps values to a bounded range for stability. Gradients propagate through β_i , letting the system learn where to relax or tighten the KL leash.

Training begins with $\kappa_{\max} = 0$ (pure imitation). Every 500 steps we evaluate validation reward; if reward stagnates and mean $\text{KL} \leq 0.8 \kappa_{\max}$, we enlarge $\kappa_{\max}$ multiplicatively by 1.5. Because the dual update enforces $\text{KL} \leq \kappa_{\max}$ at every step, safety is maintained across expansions.

4 Method

Each optimisation iteration executes seven operations.

1. Feature extraction: For every preference pair, compute input length, reference margin, reward-model entropy, and dataset ID. Detach features from gradients.
2. β prediction: Pass features through a two-layer MLP (hidden 16, tanh activation, softplus output) to obtain β_i ; clamp to a bounded range.
3. Loss computation: For each pair compute ρ_i and the scaled logistic loss $-\log \sigma(\beta_i \rho_i)$. Average across the batch.
4. Primal update: Back-propagate through both policy and MLP, then update parameters with AdamW (learning rate 1×10^{-5} , cosine decay, gradient accumulation 4, mixed precision).
5. KL estimate: Compute KL_{batch} as the mean divergence between π and π_0 over chosen and rejected responses.
6. Dual update: Adjust β_{dual} via $\beta_{\text{dual}} \leftarrow \max(0, \beta_{\text{dual}} + \eta (\text{KL}_{\text{batch}} - \kappa_{\max}))$ with $\eta = 0.05$.
7. κ management: On validation checkpoints, grow $\kappa_{\max}$ by 1.5 if reward has plateaued and $\text{KL} \leq 0.8 \kappa_{\max}$.

134 Substituting Step 3 with DiscoPOP’s blended loss or any pairwise surrogate that exposes a temperature
 135 yields an equally self-tuning algorithm. The implementation adds roughly forty lines to a standard
 136 DPO loop.

137 Because β_{dual} follows projected stochastic ascent on a convex dual function and the primal step uses
 138 unbiased gradients, the running average of constraint violations converges to zero. Across all tasks
 139 and seeds our logs record no KL breaches.

Algorithm 1 C3PO training loop

Inputs: dataset of preference pairs $\mathcal{D}$; reference policy π_0 ; policy parameters θ ; MLP parameters ϕ ; initial $\kappa_{\text{max}} \leftarrow 0$; dual multiplier $\beta_{\text{dual}} \leftarrow 0$; step size η

for each training step **do**

Sample mini-batch $\mathcal{B} \subset \mathcal{D}$

Extract features x_i from reference-only signals for each pair $i \in \mathcal{B}$

Predict temperatures: $\beta_i \leftarrow \text{softplus}(\text{MLP}_{\phi}(x_i))$; clamp to bounded range

Compute margins ρ_i and loss $\ell_i \leftarrow -\log \sigma(\beta_i \rho_i)$

Average loss $\mathcal{L}_{\text{primal}} \leftarrow |\mathcal{B}|^{-1} \sum_{i \in \mathcal{B}} \ell_i$

Update θ, ϕ by descending $\nabla_{\theta, \phi} \mathcal{L}_{\text{primal}}$ (AdamW)

Estimate $\text{KL}_{\text{batch}} \leftarrow |\mathcal{B}|^{-1} \sum_{i \in \mathcal{B}} \text{KL}(\pi_{\theta}(\cdot | i) \| \pi_0(\cdot | i))$

Update dual: $\beta_{\text{dual}} \leftarrow \max(0, \beta_{\text{dual}} + \eta (\text{KL}_{\text{batch}} - \kappa_{\text{max}}))$

if validation checkpoint **then**

if reward plateaued **and** mean $\text{KL} \leq 0.8 \kappa_{\text{max}}$ **then**

$\kappa_{\text{max}} \leftarrow 1.5 \kappa_{\text{max}}$

end if

end if

end for

140 5 Experimental Setup

141 5.1 Models and datasets

142 We fine-tune Pythia-1.4 B and Mistral-7 B on five tasks: AlpacaFarm and UltraFeedback dia-
 143 logue, Reddit TL;DR summarisation, IMDb Positive steering, and Anthropic Harmless safety. Pre-
 144 processing deduplicates prompts, filters lengths 4-512, discards low-margin pairs ($|\Delta R| < 0.05$),
 145 tokenises to 512 tokens, and pads to multiples of eight. The 80/10/10 split (seed 2023) yields 24 621
 146 UltraFeedback-HF, 23 315 UltraFeedback-Argilla, 40 851 Harmless, and 10 154 MedQA pairs -
 147 matching protocol counts.

148 5.2 Training protocol

149 All methods share optimiser settings: AdamW (learning rate 1×10^{-5} , $\beta_1 = 0.9$, $\beta_2 = 0.95$,
 150 weight decay 0.1), cosine schedule with 200 warm-up steps, three epochs, batch 128 preference pairs,
 151 gradient accumulation 4, fp16 forward with bfloat16 accumulation. Each experiment uses seeds {42,
 152 1234, 2023}; we report mean $\pm$ 95% confidence intervals.

153 5.3 Baselines

- 154 • DPO grid $\beta \in \{0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 6.4, 10\}$
- 155 • DiscoPOP grid $\beta \in \{0.05, 0.1, 0.2, 0.5, 1, 2, 3, 5\}$
- 156 • PPO grid $\kappa \in \{0.5, 1, 2, 5\}$
- 157 • SA-DPO with published PID gains

158 5.4 Metrics

159 Primary metrics are (1) normalised reward and (2) mean KL on the validation split. Secondary
 160 metrics include ROUGE-L (TL;DR), sentiment accuracy (IMDb-P), MT-Bench overall (dialogue),
 161 harmlessness rate (Anthropic classifier), GPU hours, wall-clock per 1k steps, energy via nvidia-smi,

162 and peak VRAM. Robustness is assessed by adding 5% Gaussian noise to rewards. Hyper-parameter
163 sensitivity is measured by sweeping $\eta \in \{0.02, 0.05, 0.1\}$ on a quarter-sized AlpacaFarm subset.

164 5.5 Specialised experiments

165 (1) Context awareness: fine-tune Mistral-7 B on Harmless, log β_i and toxicity, and evaluate on benign,
166 entropy-listed, jail-break, and medical splits. Freeze the β -network after one epoch to gauge impact.
167 (2) Stability: train Pythia-1.4 B on UltraFeedback for one million steps and multiply rewards by 0.3
168 between steps 3 000-3 500. Compare C3PO, SA-DPO, and fixed- β DPO while logging KL - $\kappa_{\max}$,
169 mean β , and reward every 50 steps.

170 6 Results

171 6.1 One-shot versus exhaustive grids

172 Averaged across five tasks and two model sizes (three seeds), C3PO attains normalised reward
173 1.00 ± 0.00 while never exceeding $\kappa_{\max}$. Best-grid DPO reaches 0.98 ± 0.01 , DiscoPOP 0.97 ± 0.02 ,
174 SA-DPO 0.96 ± 0.02 with 7% constraint violations, and PPO 0.92 ± 0.03 with 24% violations. GPU
175 hours per task drop from 48 ± 3.1 (DPO grid) and 40 ± 2.7 (DiscoPOP grid) to 8 ± 0.4 (C3PO).
176 Mean KL stays within 5% of $\kappa_{\max}$ throughout, empirically validating trust-region control.

177 6.2 Context awareness and safety

178 Median β_i adapts to risk: 0.7 on benign prompts, 1.2 on long or high-entropy prompts, 2.5 on medical
179 domain queries, and 3.8 on jail-break prompts. Spearman correlations are $\rho(\beta_i, \text{toxicity}) = 0.62$
180 and $\rho(\beta_i, \text{RM-entropy}) = 0.48$. Freezing the β -network increases KL variance by 33% and drops
181 harmlessness by 7%, demonstrating that learned regularisation is functionally important. Removing
182 20% toxic samples leaves β_i on jail-break prompts high, indicating robustness.

183 6.3 Stability under reward shocks

184 During the synthetic $0.3\times$ reward perturbation, C3PO logs zero KL violations and recovers 95% of
185 the pre-shock reward in 380 steps. SA-DPO spikes to $1.4 \kappa_{\max}$ and needs over 900 steps to recover;
186 fixed- β DPO diverges. The KL oscillation standard deviation over 1 000-step windows is 0.07 (C3PO)
187 versus 0.31 (SA-DPO) and 0.46 (fixed- β).

188 6.4 Ablations

189 (1) Replacing per-sample β_i with a scalar lowers reward by 1.7% and raises KL variance 29%. (2)
190 Substituting dual ascent with proportional control reproduces SA-DPO oscillations. (3) Wrapping
191 DiscoPOP with C3PO removes its temperature sweep while matching reward. (4) Adding 5% reward
192 noise reduces final reward by only 0.4% with C3PO versus 1.3% with best-grid DPO.

193 6.5 Limitations

194 The dual step size η remains a mild hyper-parameter; values 0.02-0.1 are stable but extremes were
195 not explored. Feature engineering is minimal; richer uncertainty estimates may further improve β_i .
196 While empirical KL violations are zero, formal convergence rates for sequence models merit future
197 study.

198 7 Conclusion

199 C3PO reframes preference fine-tuning as a self-tuning, safety-aware constrained optimisation problem.
200 By embedding dual-ascent theory, a context-aware per-sample temperature, and an automatic KL
201 budget rule, it removes the dominant hyper-parameters of existing methods, achieves higher reward
202 and lower KL variance, and strictly satisfies trust-region constraints while cutting compute by
203 approximately six-fold. The wrapper is loss-agnostic, scales to seven-billion-parameter models on
204 commodity hardware, and complements advances in surrogate-loss design Lu et al. [2024], Shao

205 et al. [2024] and analyses of mode-seeking behaviour Tajwar et al. [2024]. Future work will extend β
206 prediction to the token level, incorporate richer uncertainty features, and establish tighter convergence
207 rates for dual ascent in high-dimensional sequence spaces.

208 **References**

- 209 Chris Lu, Samuel Holt, Claudio Fanconi, Alex J. Chan, Jakob Foerster, Mihaela van der Schaar, and
210 Robert Tjarko Lange. Discovering preference optimization algorithms with and for large language
211 models. 2024.
- 212 Chenze Shao, Fandong Meng, Yijin Liu, and Jie Zhou. Language generation with strictly proper
213 scoring rules. 2024.
- 214 Fahim Tajwar, Anikait Singh, Archit Sharma, Rafael Rafailov, Jeff Schneider, Tengyang Xie, Stefano
215 Ermon, Chelsea Finn, and Aviral Kumar. Preference fine-tuning of llms should leverage suboptimal,
216 on-policy data. 2024.